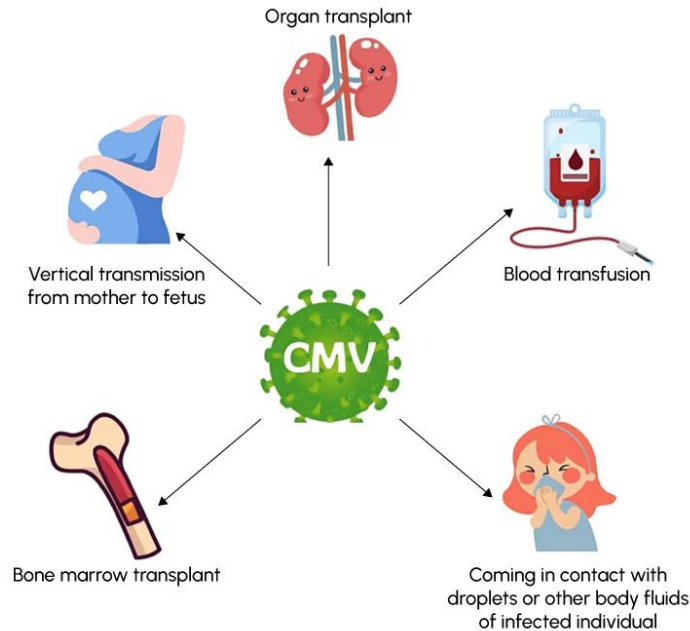




Analysis of Cytomegalovirus-Infected Human Blood Genes

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Motivation + Background



Cytomegalovirus (CMV)

A common virus that leaves detectable signals in blood gene expression

Questions

- 1) Can blood gene expression be used to classify cell identity?
- 2) Can blood gene expression be used to predict ethnicity group?
- 3) Can blood gene expression be used to distinguish CMV-related disease status?

The Data



Filtering Pipeline

717,559 CD8 T cells → 310,732 cells retained (43.3%)

- Remove unknown ethnicity: -382,049
- Keep CMV+/CMV- Only: -3,128
- Gene Count $1,000 \leq n_{\text{genes}} \leq 3,900$: - 5289
- Doublet score < 0.1 : -11,119
- Mitochondrial Content $\leq 3.8\%$: -4,450
- Remove rare cell types ($< 4,000$ cells): -792

Total After Filtering: 310,732

Cells Retained: 43%

Normalization & HVG Selection

Counts scaled to 10,000 per cell, log1p variance stabilization.
Seurat HVG selection → top 2,000 genes.

6 Cell Types

- Effector memory CD8+ α - β T cell
- Naive thymus-derived CD8+ α - β T cell
- Mature γ - δ T cell
- Mucosal invariant T cell (MAIT)
- Central memory CD8+ α - β T cell
- Double negative thymocyte

Donor-Based Split (70 / 15 / 15)

Train: 198,974 (70%)

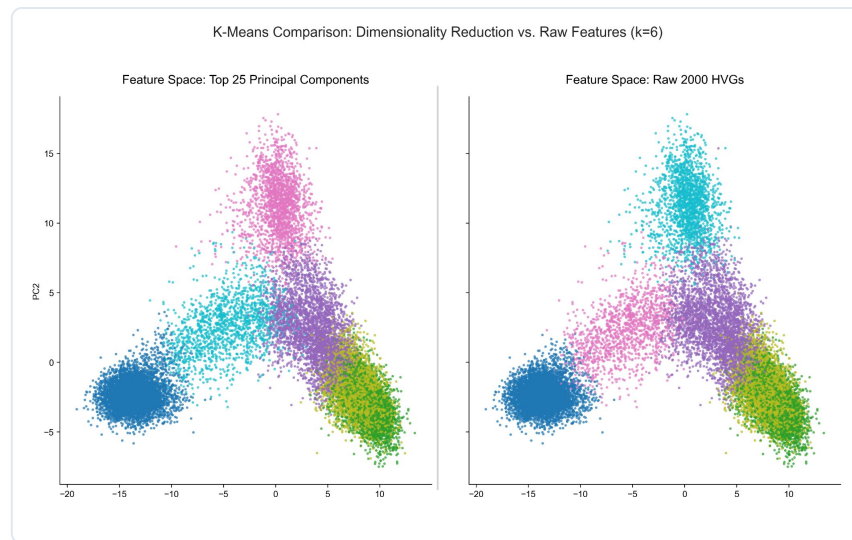
Val: 53,612 (15%)

Test: 58,146 (15%)

Clustering Analysis

Approach

- **Goal:** Verify 2,000 HVGs capture cell-type structure
- **Parameters:** $K = 6$ (one per known cell type)
- **Sample:** 20,000 cells x full 2,000 gene space



Clustering Analysis

Key Insights

- Naïve CD8+ T cells cluster with perfect purity (100%)
- MAIT cells and double negative thymocytes are also well-resolved (95%, 87%)
- Effector memory CD8+ T cells are the most poorly resolved — scattered across 3 clusters
- $\gamma\delta$ T cells disperse broadly, reflecting inherent subset diversity

Conclusion: HVGs are biologically informative



XGBoost

Approach

Goal:

Predict ethnicity based on gene expression patterns

Rationale:

- Handle high dimensional data
- Capture complex genetic relationships

Key Insights

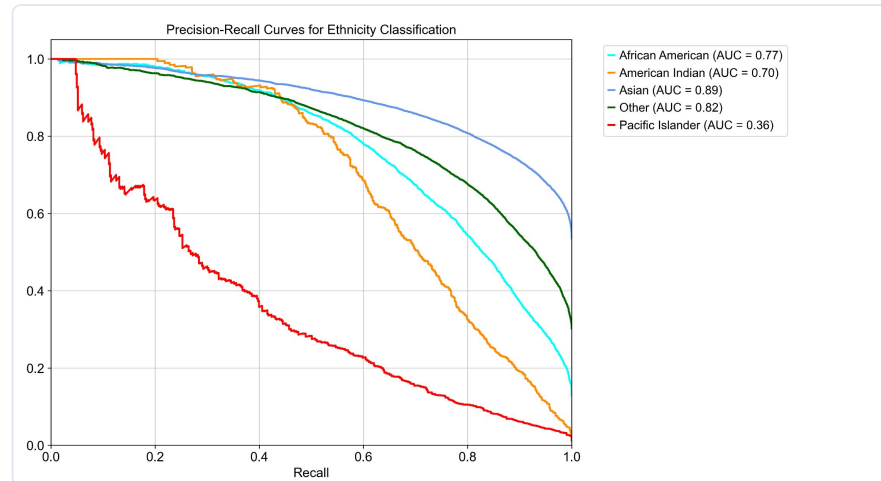
Accuracy: **77.81%**

Recall: **77.81%**

Precision (Weighted): **78.19%**

ROC-AUC (One-vs-Rest): **93.11%**

ROC-AUC (One-vs-One): **92.17%**



Logistic Regression



Methodology

Goal:

Evaluate whether incorporating ethnicity information improves CMV classification.

Approach:

- Logistic Regression model
- Lasso-based feature selection (2000 → 1898 genes)
- Decision Threshold: 0.65
 - Class 0: 25,989
 - Class 1: 27,623

Baseline: No Prior Information

Precision: **60.5%**
Recall: **60.4%**
F1 Score: **60.38%**
ROC-AUC: **66.52%**

With Ethnicity Prior Information:

F1 Score improved to **68.22%**

Alternative: Pseudo-bulking Method

Result of **53%** - *deemed insufficient performance*

Thats all folks!



Conclusion: CD8 T, $\gamma\delta$ T, MAIT, and dnT cells provide significantly higher informational value compared to other cell types for our primary tasks.

Any Questions?

